

Econ 30065: Microeconomics

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C5

Preferences

- **Preferences** tell us about a consumer's likes and dislikes
- (Assumptions) Rational consumers follow two basic decision-making principles:
 - **Ranking principle**
 - **Choice principle**
- (Assumption) **The more is better Principle**

Principles of Decision making

- **Ranking principle:** a consumer can rank, in order of preference (though possibly with ties), all potentially available consumption bundles

Implications

- Preferences are **complete**: between any pair of alternatives, a consumer either prefers one to the other or is indifferent between them.
 - A consumer is **indifferent** between two alternatives if he likes (or dislikes) them equally
- Preferences are **transitive**: if the consumer prefers one alternative to a second, and prefers the second alternative to a third, then he also prefers the first alternative to the third.

Principles of Decision Making

- **The choice principle:** among the available alternatives, the consumer selects the one that he ranks the highest:

X, Y, Z

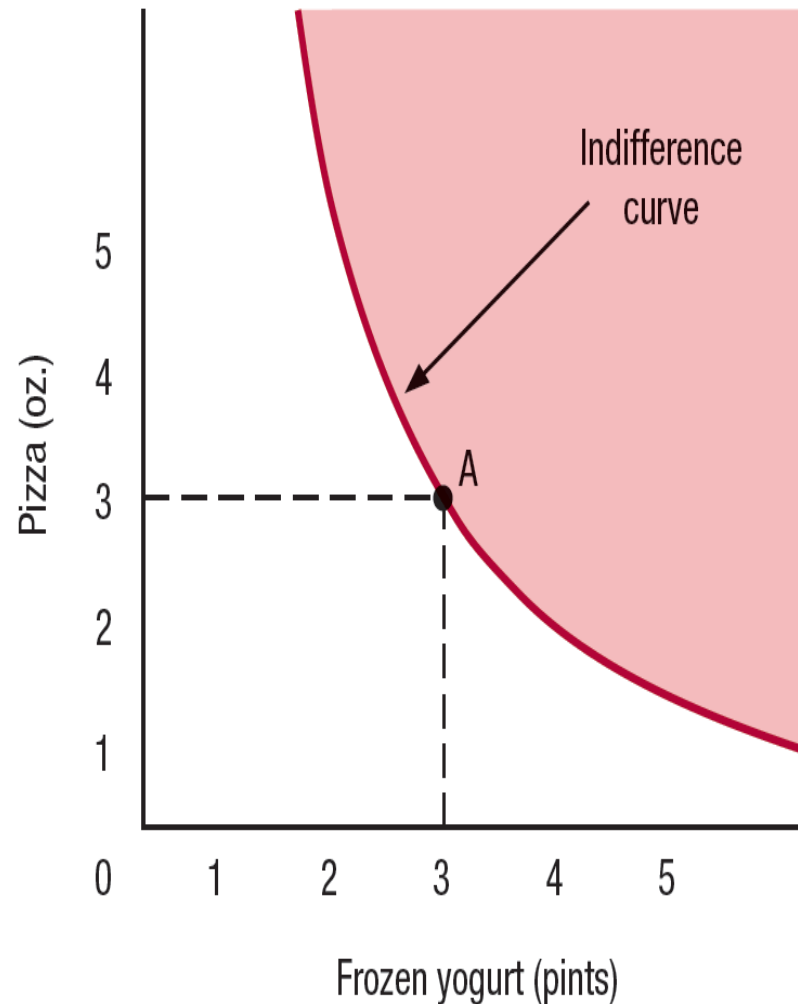
- If Z is preferred to X, and
- Z is preferred to Y, then
- Z is the selected alternative

Principles of Decision Making

- **The More-is-better Principle:** When one consumption bundle contains more of every good than a second bundle, a consumer prefers the first bundle to the second (free disposal)

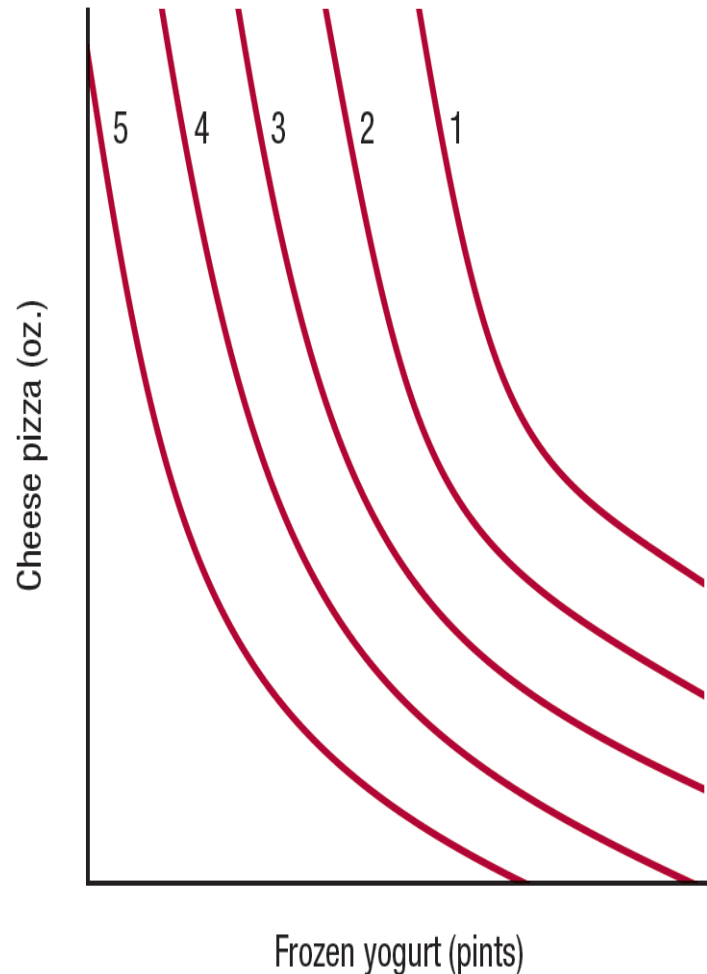
Graphical approach: Indifference curves

- Consumer preferences can be graphically represented through **indifference curves**
- An indifference curve shows all consumption bundles that a consumer likes equally well
- Math function $Y=f(x)$
- Different agents have different indifference curves



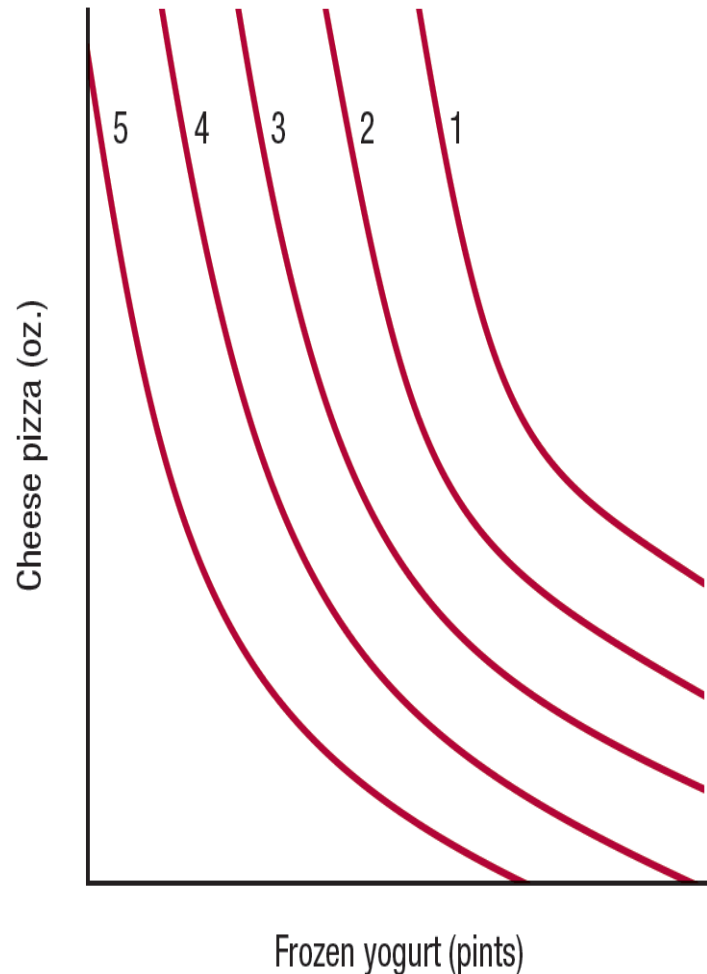
Families of Indifference Curves

- A family of **indifference curves** is a collection of indifference curves that represents one individual's preferences



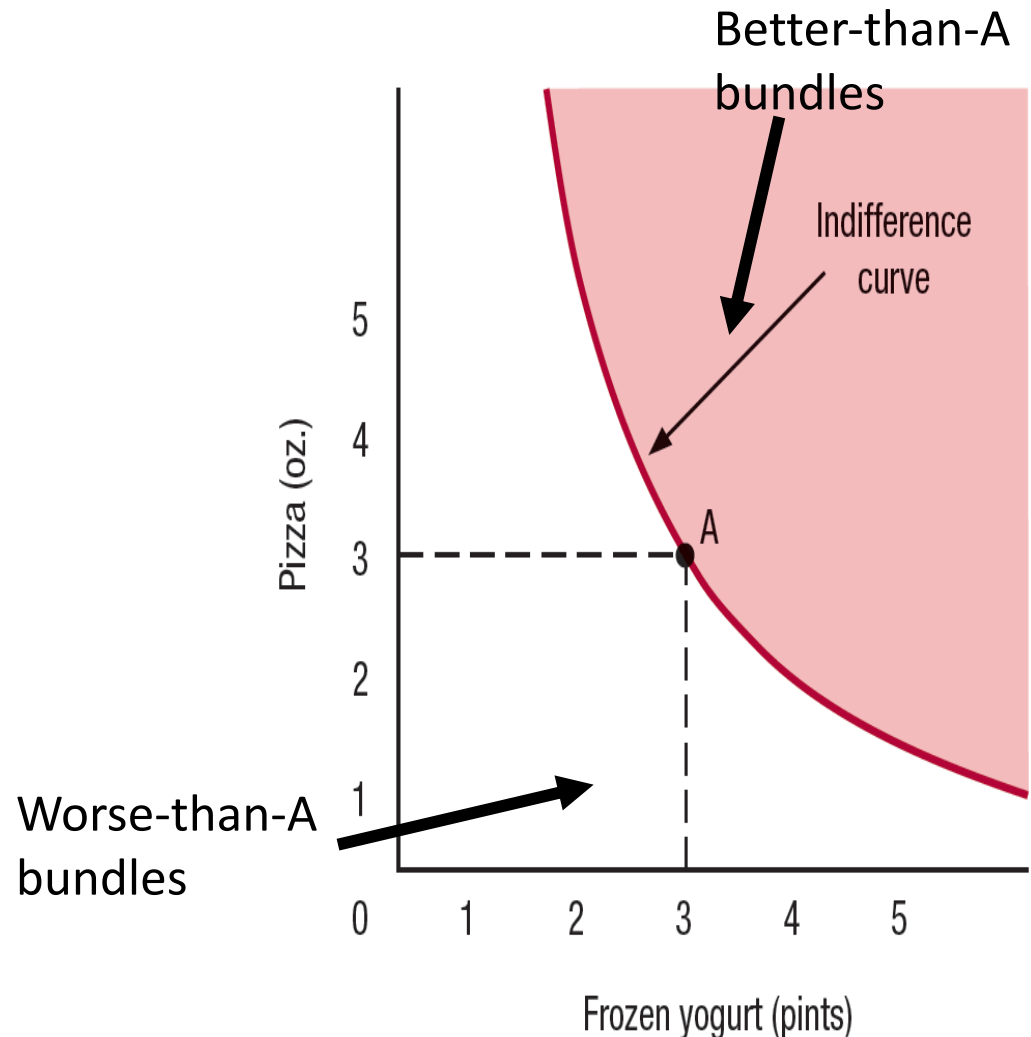
Properties of indifference curves

- Consumer prefers bundles on IC furthest from the origin (more is better principle)



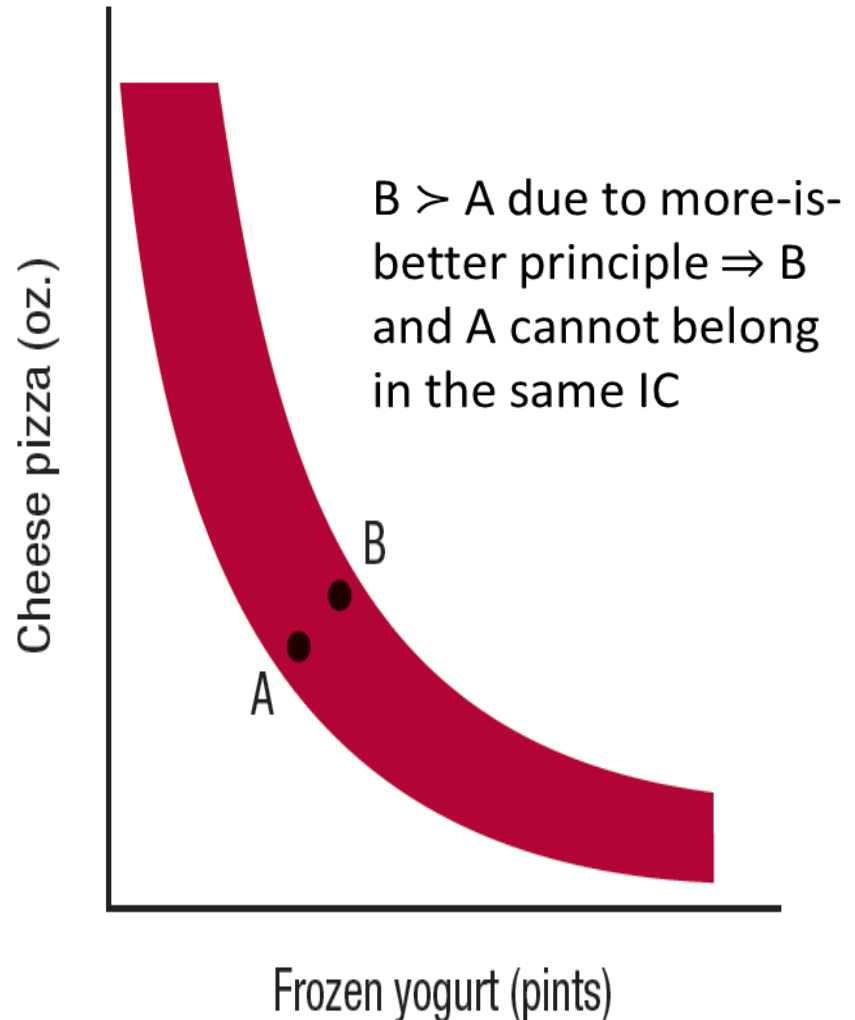
Properties of indifference curves

- The IC that runs through consumption bundle A separates all the better-than-A bundles from the worse-than-A bundles



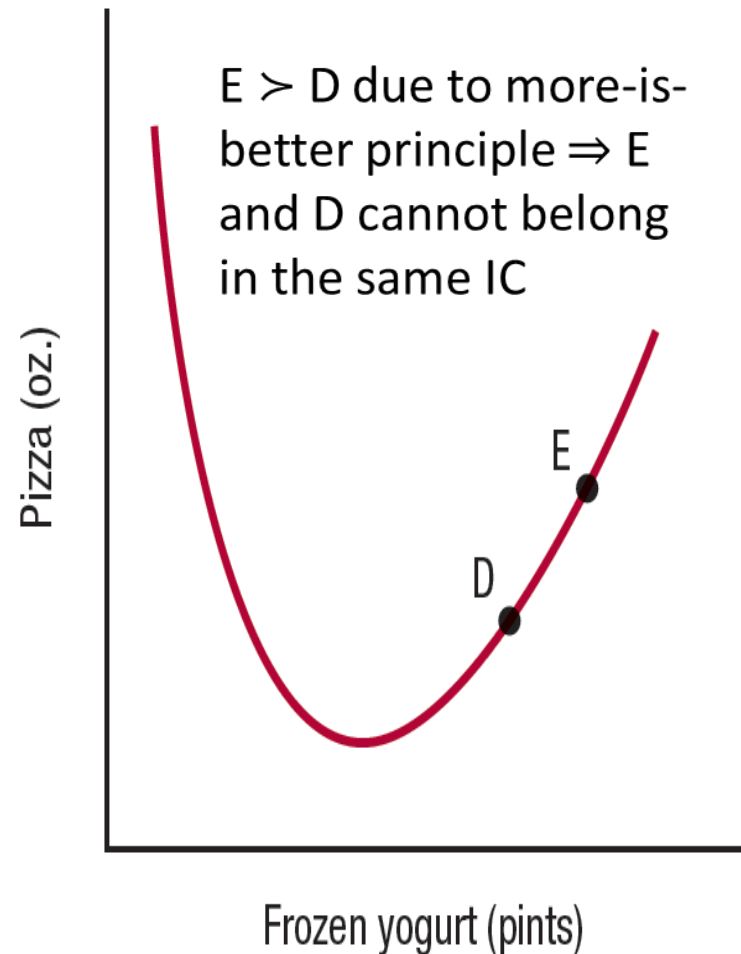
Properties of indifference curves

- Indifference curves are thin



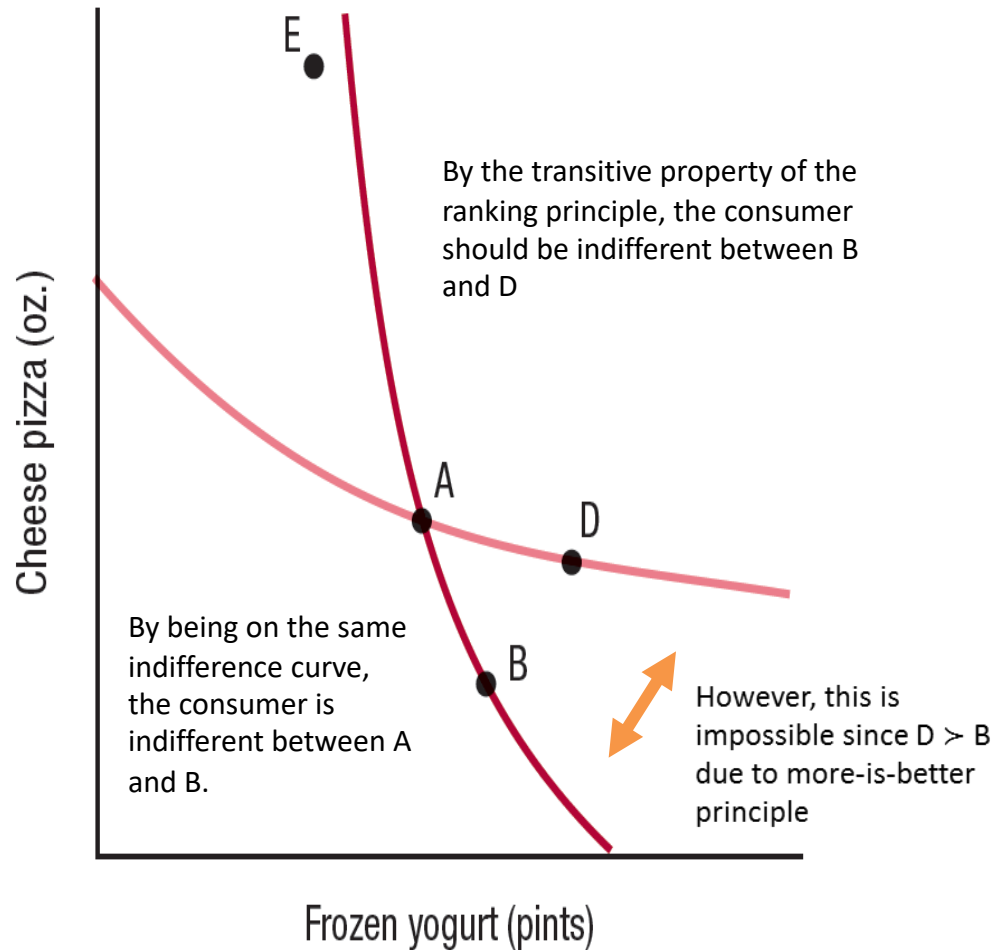
Properties of indifference curves

- Indifference Curves do not slope upward



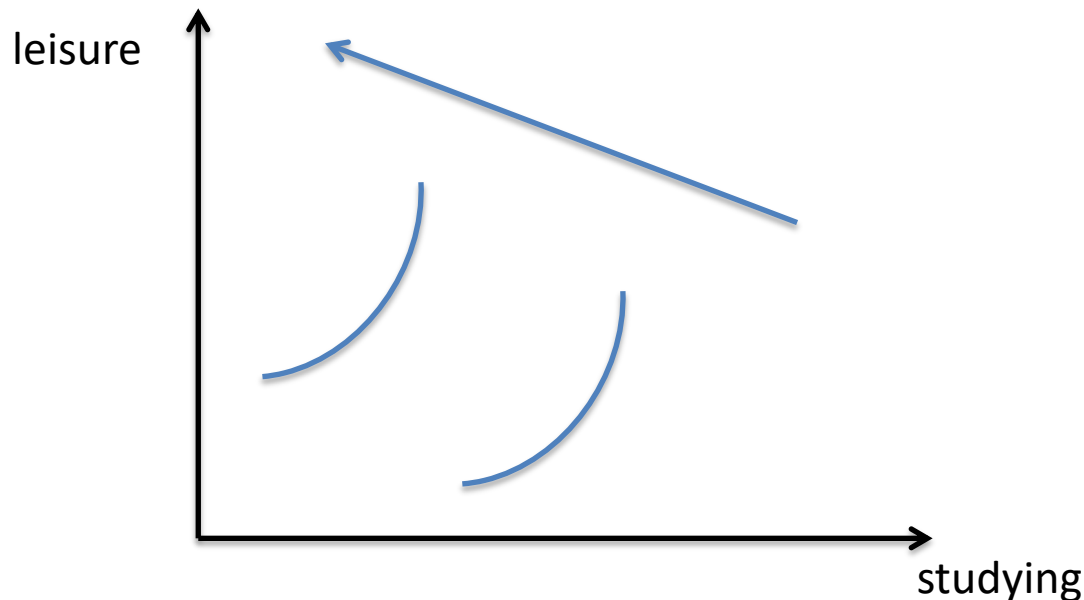
Properties of indifference curves

- IC curves from the same family do not cross



Bads vs. Goods

- A bad is a product that makes a consumer worse off: less is better (think at Pollution)!
- Example: hours of studying vs. hours of leisure



Substitution between goods

- The IC contains information about the rate at which a consumer can substitute one good for another so that she is indifferent between the two bundles
- The rate of substitution between 2 bundles is given by $-\Delta Y/\Delta X$

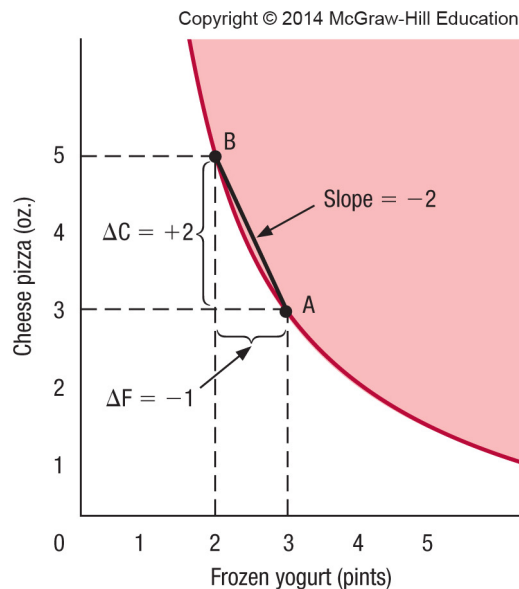


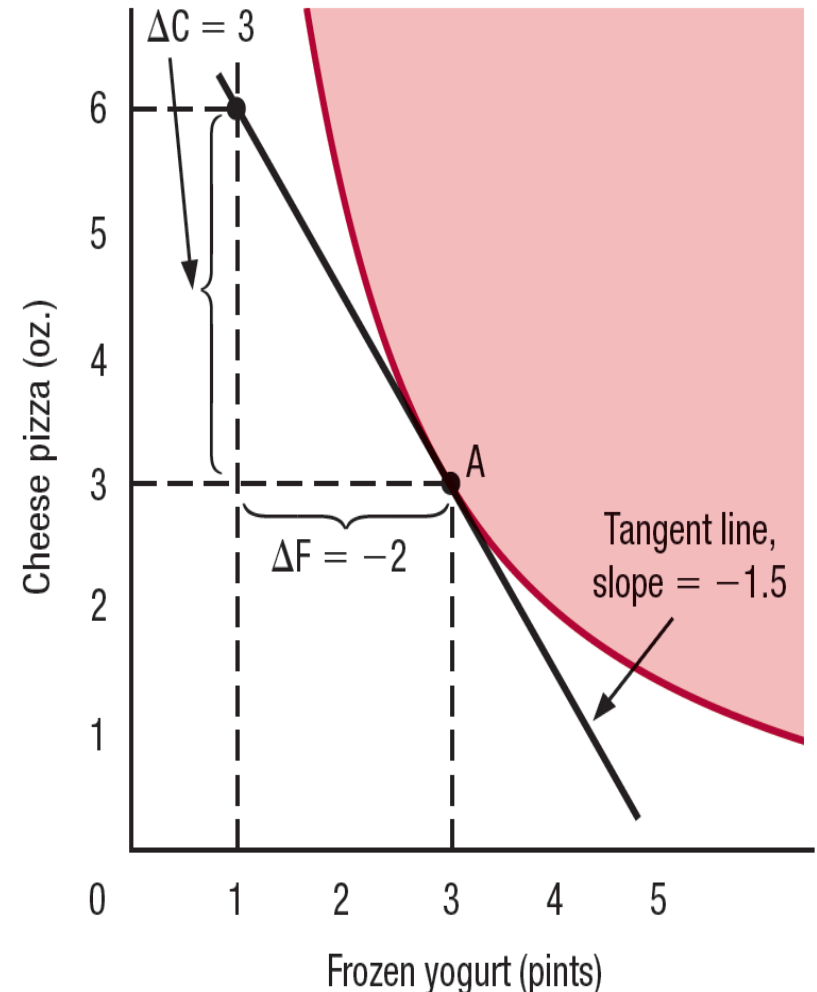
Figure 4.7

Indifference Curves and Rates of Substitution.

In moving from bundle A to bundle B, Madeline loses 1 pint of yogurt and gains 2 ounces of pizza. So the rate at which she is willing to substitute for yogurt with pizza is 2 ounces per pint.

Marginal rate of substitution between goods

- The rate at which Madeline is willing to substitute small amounts of pizza with yogurt is called **marginal rate of substitution for yoghurt with pizza**
- The marginal rate of substitution at bundle A equals the (absolute value) of the slope of the line drawn tangent to the indifference curve running through point A.

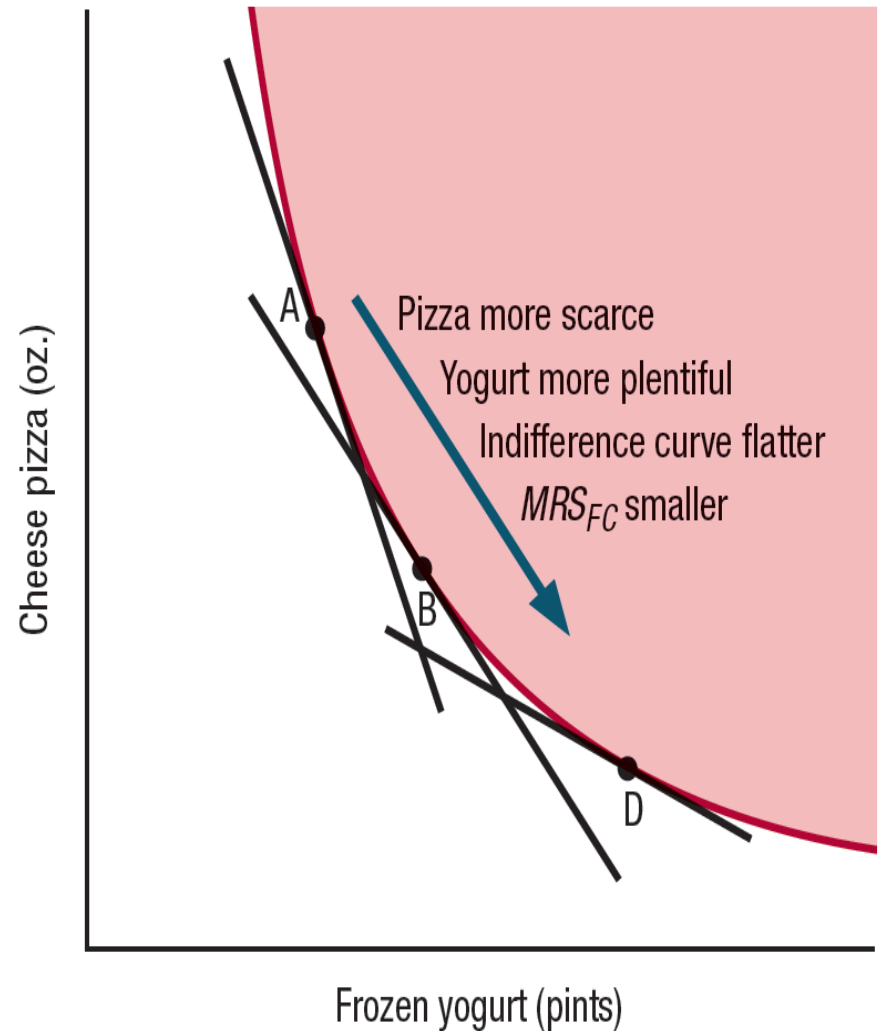


Substitution between goods

Marginal Rate of Substitution for X with Y (at a given point): The amount of good Y that a consumer is willing to give up for an extra unit of good X, such that the agent is on the same indifference curve as before.

MRS characteristics

- Assume convex indifference curves (preference for variety)
- Within the same indifference curve, as pizza becomes more scarce and yogurt becomes more plentiful, the MRS for yogurt with pizza falls



MRS characteristics

- Angie likes yogurt relative to pizza more than Marcus
- Angie's MRS for yogurt with pizza is higher than Marcus' MRS

